

Testing:

In the “tests” package there are four classes designed to test the functionality of the Account, CardDeck, Table, and Message classes. Some of the classes are dependent on the “JUnitTest.txt” file, which is included in the Multiplayer-Blackjack project. The last class in the package is TestSuite, which runs all tests simultaneously. A test runner is not included as JUnit 6 was used to create these tests and they are not available in that version.

1. CardDeck (Tests for CarDeck class)
 - a. testCreatingOneDeck()
 - i. Tests the constructor and pullCard method to ensure a single deck initializes with exactly 52 cards, containing every valid combination of rank and suit without duplicates or missing cards.
 - b. testShuffleChangesOrder()
 - i. Tests the shuffle method by pulling all cards from a shuffled deck and comparing their order to an unshuffled deck, verifying that the cards are randomized while maintaining the correct total count.
 - c. testResetRestoresFullDeck
 - i. Tests the reset method by simulating a game (drawing a subset of cards) and verifying that calling reset completely repopulates the shoe back to its maximum capacity (e.g., 104 cards for a 2-deck shoe).
 - d. testAddMaxDecksIncreasesCapacity()
 - i. Tests the addMaxDecks method to verify that it successfully increments the shoe's capacity and repopulates the deck (for example: expanding a 1-deck shoe of 52 cards into a 2-deck shoe of 104 cards).
2. Table (Tests for Table Class)
 - a. drawCardTest()
 - i. Tests the drawCard method by starting a game and verifying it successfully returns a non-null Card object from the table's shoe.
 - b. testAddUserToTable()
 - i. Tests the addUserToTable method to ensure a user with dealer credentials is assigned as the dealer, and subsequent users are seated and counted correctly as players.
 - c. testRemovePlayerFromTable()
 - i. Tests the queueLeave method to verify that standard players are removed from their seats and the player count decreases immediately when requested.
 - d. testNextTurnSkipsBustedPlayers()
 - i. Tests the playerStoodOrBust and nextTurn methods to ensure the game loop correctly bypasses player who have already stood or busted.
 - e. testStartGameDealsCards()
 - i. Tests the startGame method to verify that it successfully enters the dealing phase and pulls exactly two cards for the active dealer and each seated player.
 - f. testDealerQueueLeaveDuringGame()

- i. Test the queueLeave and endGame methods to verify the requirement that a dealer attempting to leave during an active game is queued and not removed until the round officially ends.
 - g. testGetVacancies()
 - i. Tests the getVacancies method to verify it returns an accurate boolean array reflecting the exact populated (false) and empty (true) seats at a 5-seat table.
 - h. testEndGameClearsDrawnCards()
 - i. Tests the endGame method to verify that the table's tracked cards vector is completely cleared and reset for the next round
- 3. Message (Tests for Message class)
 - a. defaultConstructorTest()
 - i. Tests Message default constructor to ensure correct default values are assigned.
 - b. twoArgConstructorTest()
 - i. Tests Message constructor that takes message type and message status as arguments to ensure that a message object's type and status are initialized with passed type and status values.
 - c. threeArgConstructorTest()
 - i. Tests Message constructor that takes message type, message status, and text as arguments to ensure that a message object's type, status, and text are initialized with passed type, status, and text values.
 - d. emptyStringTest()
 - i. Tests that Message constructor properly assigns empty string text values.
 - e. nullStringTest()
 - i. Tests that Message constructor properly assigns null text values.